

Gauge Invariance Obstructs R-Charge Concentration in Fermionic Matrix Models

Abstract. We study a three-parameter family of purely fermionic gauge quantum mechanics: p flavours of adjoint fermions for a rank- r gauge algebra, with a nilpotent supercharge of odd degree q . The BPS states are the cohomology of Q , which acts on the Fock space as multiplication by a fixed q -form $\omega \in \Lambda^q V$. We find that the BPS states occupy a window of consecutive R-charges whose width grows *linearly in the rank*, $W = r(w_s - 1) + 1$ at the maximal weight and $W \geq r(2 \min(p, q - 1) - p) + 1$ at every weight for $u(N)$. Since the irreducible complexes are graded by $N_\Psi \bmod q$, a window wider than q cannot place one degree per class, and R-charge concentration fails for all r beyond a fixed bound. The mechanism is identifiable: gauge invariance forces every interaction term to carry zero weight, and the positive-root subspace — half of any reductive Lie algebra — admits no such term. The resulting interaction hypergraph is sparse, with transversal density $\tau/n \rightarrow \frac{1}{2}$, whereas ungauged SYK has $\tau/n \rightarrow 1$ and is unobstructed. The obstruction is therefore attributable to gauge invariance itself, not to the flavour count, the choice of couplings, or the loss of Casimir solvability. Fortuity, by contrast, responds to gauge invariance in the *opposite* way: the stabilisation map $H_N \rightarrow H_{N+1}$ vanishes identically, because embedding creates a new site whose self-loop is necessarily unoccupied. All statements are established for $u(N)$ and verified against exact integer computations for $N \leq 5$; the $\mathfrak{su}(N)$ case is discussed and left open.

1 The models

Let $\mathfrak{g}$ be a reductive Lie algebra of rank r with a fixed invariant form, and let Ψ^a , $a = 1, \dots, p$, be complex fermionic fields valued in $\mathfrak{g}$. For $\mathfrak{g} = \mathfrak{g}(N)$ we write Ψ_{ij}^a with $i, j = 1, \dots, N$ and

$$\{\Psi_{ij}^a, \bar{\Psi}_{kl}^b\} = \delta^{ab} \delta_{il} \delta_{jk}, \quad (1)$$

the gauge group acting by conjugation. The Fock space is the exterior algebra

$$\mathcal{H} = \Lambda^\bullet V, \quad V = \mathbb{C}^p \otimes \mathfrak{g}, \quad n \equiv \dim V = p \dim \mathfrak{g}, \quad (2)$$

graded by the R-charge $k = N_\Psi = \text{Tr} \Psi^a \bar{\Psi}^a$, so that $\Lambda^k V$ is the k -fermion sector, of dimension $\binom{n}{k}$. We call a basis vector of V a *mode*; a basis vector of $\Lambda^k V$ is then a set S of k modes, and we write $m_S = \prod_{v \in S} v$ for the corresponding state. The wedge product is the anticommuting product, so that $v \wedge v = 0$ is Pauli exclusion written algebraically. All dynamics comes from a single supercharge,

$$Q = \sum_{a_1 \dots a_q} C_{a_1 \dots a_q} \text{Tr}[\Psi^{a_1} \dots \Psi^{a_q}], \quad H = \{Q, \bar{Q}\} \geq 0, \quad [N_\Psi, Q] = qQ. \quad (3)$$

Only *odd* q is admissible, and for two independent reasons. A cyclic shift inside $\text{Tr}[\Psi^q]$ costs $(-1)^{q-1}$, so the trace vanishes identically for even q ; and $Q^2 = 0$ requires that exchanging the two q -tuples in $C \otimes C$ — a permutation of $2q$ fermions built from q^2 transpositions, contracted against a totally antisymmetric product — produce a relative minus sign, which again happens precisely for odd q . The case $(p, q) = (1, 3)$ is Chen’s exactly solvable single-matrix model, in which $H = 3N(N^2 - 1) - 9\hat{C}_2$ is a function of the gauge Casimir alone; $(p, q) = (3, 3)$ is his proposed “matrix SYK” model, for which that Casimir structure is lost.

The structural fact that organises everything below is that Q is multiplication by a form. Every Ψ is a creation operator, so Q contains no annihilators whatsoever and acts on $\Lambda^\bullet V$ as nothing more than left multiplication:

$$Q = \omega \wedge -, \quad \omega = \sum C_{a_1 \dots a_q} \text{Tr}[\Psi^{a_1} \dots \Psi^{a_q}] \in \Lambda^q V, \quad H^\bullet = \text{Ann}(\omega)/(\omega). \quad (4)$$

For $p = 1$ the form $\text{Tr} \Psi^3$ is proportional to the Cartan 3-form $\text{Tr}(x[y, z])$ of $\mathfrak{g}$. The entire problem is thus the cohomology of multiplication by one specific, highly non-generic element of $\Lambda^q V$, and the question of R-charge concentration becomes a question about where in the exterior algebra that multiplication fails to be injective or surjective.

The Cartan torus acts on Ψ_{ij} by $u_i u_j^{-1}$, so that mode carries weight $e_i - e_j$ and a state’s weight is the sum over its modes, i.e. the row sums minus the column sums of the occupation matrix n_{ij} . Diagonal modes carry

(a) maximal weight freezes the off-diagonal

	$j=1$	$j=2$	$j=3$
$i=1$	free	p	p
$i=2$	0	free	p
$i=3$	0	0	free

occupation n_{ij} at λ^*

(b) only self-loops can close a cycle

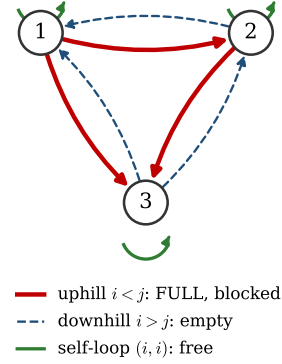


Figure 1: (a) At the maximal weight the occupation is forced: every mode with $i < j$ is full, every mode with $i > j$ is empty, and only the diagonal is free. (b) Reading modes as coloured directed edges, Q creates a closed walk $i \rightarrow j \rightarrow k \rightarrow i$ and needs all its edges empty. Uphill edges are occupied, so every creatable step is non-increasing; a cycle of non-increasing steps must be constant, leaving only self-loops.

$e_i - e_i = 0$ and are invisible to the weight. Because Q is a gauge singlet it preserves the weight, and because $[N_\Psi, Q] = qQ$ it raises k by q ; the cohomology therefore splits into irreducible complexes labelled by

$$(\lambda, \omega_{\mathbb{H}}, c), \quad \lambda \in \widehat{U(N)}, \quad \omega_{\mathbb{H}} \in \mathbb{Z}_p, \quad c = k \bmod q. \quad (5)$$

R-charge concentration, in the sense of Chang, Chen, Sia and Yang, is the statement that in each complex of macroscopic index all BPS states sit at a single degree. Writing W for the number of consecutive R-charges that carry BPS states, a necessary condition is therefore

$$\boxed{W \leq q} \quad (6)$$

since a window wider than the grading period must, by pigeonhole, place two occupied degrees in one residue class. Everything that follows is an estimate of W .

2 The maximal irrep: index loops and the slot factorisation

2.1 The maximal weight

The highest weight appearing in the Fock space is found greedily: fill row 1 and empty column 1, then do the same with row 2, and so on. The result is the fully upper-triangular occupation, with

$$\lambda_i^* = p(N+1-2i), \quad C_2^{\max} = \frac{1}{2}p(p+1) \sum_i (N+1-2i)^2 = \frac{p(p+1)N(N^2-1)}{6}, \quad (7)$$

where we used $\lambda_i^* + N+1-2i = (p+1)(N+1-2i)$ together with $\sum_i (N+1-2i)^2 = N(N^2-1)/3$. At $p=1$ this reduces to $N(N^2-1)/3$, which is Chen's $C_2(r_*)$, so (7) is its multi-flavour generalisation; at $p=3$ it gives $2N(N^2-1) = 48, 120, 240$ for $N=3, 4, 5$. Shifting λ^* to a partition gives row lengths $p(2N-2i)$ — exactly p times Chen's staircase, with the same $N-1$ non-empty rows. Adding flavours stretches the diagram sideways without adding a single row, a point we return to in §5.

2.2 Index loops

At λ^* the off-diagonal sector is completely rigid — every upper mode occupied, every lower mode empty (Fig. 1a) — so the weight space is just an exterior algebra on the zero-weight modes,

$$(\Lambda V)_{\lambda^*} \cong \Lambda^\bullet(\mathcal{Z}), \quad \mathcal{Z} = \text{Cartan} \otimes \mathbb{C}^p, \quad \dim \mathcal{Z} = pr. \quad (8)$$

What survives of Q on this space is severely restricted, and the restriction is best seen graphically. Think of a mode Ψ_{ij}^a as a directed edge $i \rightarrow j$ carrying colour a . A term of Q is a trace, so the q modes it creates form

N	window (N_Ψ)	W	BPS multiplets h^k
2	5, 6, 7	3	9, 18, 9
3	12, ..., 15	4	27, 81, 81, 27
4	22, ..., 26	5	81, 324, 486, 324, 81
5	35, ..., 40	6	243, 1215, 2430, 2430, 1215, 243

Table 1: Exact BPS multiplets of the maximal irrep at $p = q = 3$, as predicted by (10): $3^N \binom{N}{j}$ spread over $N + 1$ consecutive charges centred on $pN^2/2$, totalling 6^N . The factor $(1+t)^N$ is identical to Chen’s single-matrix Z_{BPS} — the model he identifies as *not* concentrated.

a closed walk $i_1 \rightarrow i_2 \rightarrow \dots \rightarrow i_1$, and by Pauli exclusion the term acts non-trivially only if every edge of that walk is currently empty. But at the maximal weight the empty modes are exactly those with $i > j$, together with whatever diagonal flavours happen to be free. Every step of a creatable walk therefore satisfies $i_t \geq i_{t+1}$: the walk may descend or stay level, but it can never climb, because all the uphill edges are already occupied. Closing the walk forces $i_1 \geq i_2 \geq \dots \geq i_1$, hence $i_1 = \dots = i_q$, and the q modes must be distinct and so carry q distinct flavours. **Only self-loops survive:** q diagonal modes of distinct flavours sitting at a single site. We have checked this exhaustively at $N = 3$, $p = 3$, where all 192 non-zero actions of Q on the maximal-weight space are of exactly this form.

2.3 Factorisation and the width formula

Since every surviving term sits at one site, Q splits as $Q = \sum_{i=1}^r Q_i$ with Q_i acting only inside site i ’s flavour space. The q creation operators of a self-loop all live at the same site and anticommute, so only the totally antisymmetric part of the coupling tensor survives and

$$Q_i = \omega_{\text{sl}} \wedge - \text{ on } \Lambda^\bullet \mathbb{C}^p, \quad \omega_{\text{sl}} = \text{Alt}(C) \in \Lambda^q \mathbb{C}^p. \quad (9)$$

The Q_i act on disjoint tensor factors of $\Lambda(\mathcal{Z}) = (\Lambda^\bullet \mathbb{C}^p)^{\otimes r}$, so Künneth applies and the Poincaré polynomials simply multiply:

$$\boxed{Z_{\text{max}}(t) = t^{k_0} (z_{\text{slot}}(t))^r}, \quad z_{\text{slot}}(t) = \sum_j \dim H^j(\Lambda^\bullet \mathbb{C}^p, \omega_{\text{sl}} \wedge) t^j, \quad (10)$$

with $k_0 = p|\Delta^+|$ counting the frozen off-diagonal modes. Multiplying r copies of a polynomial adds their degree-spans, so if z_{slot} is supported on w_s degrees then

$$W = r(w_s - 1) + 1. \quad (11)$$

This is the width of the BPS window at the maximal weight, in closed form, for every (p, q, N) .

It is worth seeing the machinery run once. Take $p = q = 3$ and $N = 3$. The slot algebra $\Lambda^\bullet \mathbb{C}^3$ has dimensions 1, 3, 3, 1, and $\text{Alt}(C)$ spans the one-dimensional $\Lambda^3 \mathbb{C}^3$, so $\omega_{\text{sl}} \wedge$ is non-zero only as the map $\Lambda^0 \rightarrow \Lambda^3$. Hence $H^0 = 0$ because that map is injective, $H^1 = H^2 = 3$ because the differential neither enters nor leaves those degrees, and $H^3 = 1 - 1 = 0$; that is, $z_{\text{slot}} = 3t + 3t^2 = 3t(1+t)$. Cubing and restoring the prefactor gives $Z_{\text{max}} = t^9(3t(1+t))^3 = 27t^{12} + 81t^{13} + 81t^{14} + 27t^{15}$, i.e. $h^{12..15} = 27, 81, 81, 27$, in exact agreement with the computed spectrum in Table 1.

2.4 The slot is never one degree wide

Everything now hinges on w_s , and a short argument shows it can never be 1. Grade $\Lambda^\bullet \mathbb{C}^p$ by $j \bmod q$ and let $E_c = \sum_{j \equiv c} (-1)^j \binom{p}{j}$ be the Euler characteristic of class c ; whenever $E_c \neq 0$ there must be cohomology somewhere in that class. Now $\sum_c E_c = (1-1)^p = 0$, so it is impossible for exactly one E_c to be non-zero. On the other hand, writing $\zeta = e^{2\pi i/q}$ and resolving the constraint, $E_c = \frac{1}{q} \sum_{m=1}^{q-1} \zeta^{-mc} (1 - \zeta^m)^p$ with every $1 - \zeta^m \neq 0$, so they cannot all vanish either. At least two classes must therefore carry cohomology, and

$$w_s \geq 2 \quad \implies \quad W \geq r + 1 \quad \text{at the maximal weight.} \quad (12)$$

Comparing with (6), concentration already requires $\text{rank } G \leq q - 1$ — a bound on the gauge group that is independent of the flavour content entirely.

a state S is CRITICAL iff every triangle straddles (filled = occupied mode, open = empty)

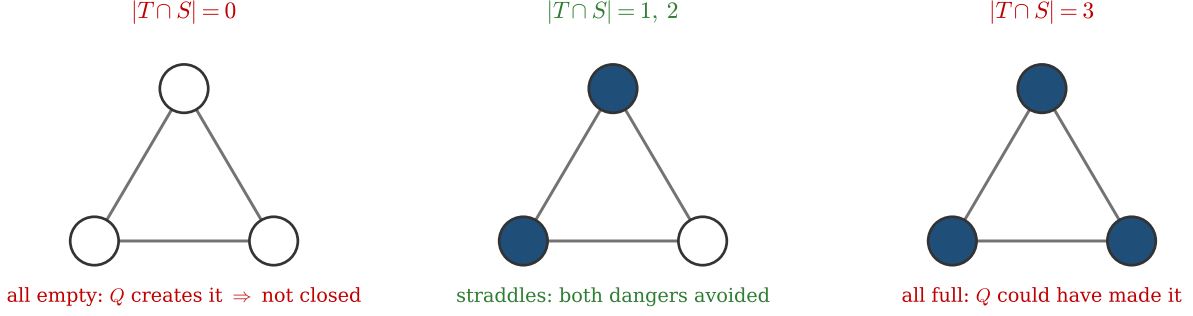


Figure 2: The three cases. If a triangle is entirely empty, Q can create it and S is not closed. If it is entirely occupied, S could have been produced by Q and the non-exactness argument fails. Only straddling triangles avoid both.

Computing z_{slot} for generic ω_{sl} at $q = 3$ gives $w_s = 2, 3, 2, 3, 4, 3, 2, 3$ for $p = 1, \dots, 8$, periodic with period four, so that

$$W = N + 1 \quad (p \equiv 3 \pmod{4}, \text{ and } p = 1), \quad 2N + 1 \quad (p \text{ even}), \quad 3N + 1 \quad (p \equiv 1 \pmod{4}, p > 1). \quad (13)$$

Two degenerations deserve comment. Whenever $q > p$ the space $\Lambda^q \mathbb{C}^p$ vanishes, the slot differential dies, and $z_{\text{slot}} = (1 + t)^p$: pushing q past the flavour count *widens* the window to its maximum rather than narrowing it. And $\dim \Lambda^3 \mathbb{C}^3 = 1$, so at $p = q = 3$ the coupling ω_{sl} is unique up to scale — Chen’s tensor is the only non-degenerate three-flavour cubic supercharge there is, and randomising C can only reproduce it or destroy it. Neither knob offers an escape.

3 Triangles and critical sets

The width formula (11) is derived at the maximal weight, where the rigidity of the off-diagonal sector does most of the work. We now give an estimate valid at *every* weight, by an elementary combinatorial argument that has the additional virtue of exposing the mechanism.

Call a *triangle* T the support — a set of q distinct modes — of a non-zero term of ω , so that $\omega = \sum_T c_T m_T$ with $c_T \neq 0$. Triangles are the elementary moves available to Q : the operator does nothing but add one. At $N = 3, p = q = 3$ there are 195 of them on 27 modes, of three kinds: 3 with all modes on one diagonal cell, 144 of the form $\{(d, d), (d, e), (e, d)\}$, and 48 with three distinct off-diagonal cells. (Only 24 of the 27 flavour triples have $C_{abc} \neq 0$, and cyclic rotations produce the same mode set.) The number to keep in mind is 195 out of $\binom{27}{3} = 2925$ possible triples: the interaction hypergraph is **sparse**, and that sparsity turns out to be the whole story.

Say that a set of modes S is *critical* if every triangle meets it partially, $0 < |T \cap S| < 3$ for all T . Equivalently, neither S nor its complement contains a triangle; equivalently again, colouring each mode by whether it lies in S , no triangle comes out monochromatic. The two inequalities do two different jobs, and Fig. 2 shows what each of them rules out.

The upper inequality makes m_S closed. For $\omega_T \wedge m_S$ to be non-zero all q modes of T must be absent from S , i.e. $T \cap S = \emptyset$, and criticality excludes exactly that; every term of ω is Pauli-blocked, so $\omega \wedge m_S = 0$. The lower inequality makes m_S non-exact. Since Q only ever adds a triangle, the only states that can reach S are those of the form $S \setminus T$ with $T \subseteq S$, so for any $y = \sum_{S'} y_{S'} m_{S'}$ the coefficient of m_S in $\omega \wedge y$ is $\sum_{T \subseteq S} \pm c_T y_{S \setminus T}$. Criticality gives $T \not\subseteq S$ for every triangle, the sum is empty, and the coefficient vanishes for every y — while m_S of course has coefficient 1 on itself. Each critical set therefore determines a non-zero class in $H^{|S|}$, and distinct critical sets give linearly independent classes, since no combination of them can be exact either.

This converts the question of the window into a purely combinatorial one: at which sizes do critical sets exist? Let α be the size of the largest triangle-free set of modes and $\tau = n - \alpha$ the transversal number of the triangle hypergraph, i.e. the size of the smallest set meeting every triangle. A set S is critical exactly when neither S nor S^c contains a triangle, which is to say that both are hitting sets, which is to say that both have size at least τ . Critical sets are thus confined to $\tau \leq |S| \leq n - \tau$, and whenever they exist throughout that range the BPS

p	τ (closed form)	τ (ILP)	bound (16)	measured W	status
1	$N(N-1)/2$	3, 6, 10, 15	$N+1$	$N+1$	tight
2	$pN(N+1)/2 - 2N$	6, 12, 20	$2N+1$	$2N+1$	tight
3	$pN(N+1)/2 - 2N$	5, 12, 22, 35	$N+1$	$N+1$	tight
4	$pN(N+1)/2 - 2N$	18	1	$2N+1$	valid, loose

Table 2: The transversal number and the resulting bound. The closed form agrees with an exact integer program in all 12 cases tested ($p = 1.4$, $N = 2.6$), and (16) is tight for $p \leq 3$. At $p = 4$ the explicit set (15) is no longer optimal and the bound degrades, though it remains valid.

cohomology is non-zero at $n - 2\tau + 1$ consecutive degrees:

$$W \geq n - 2\tau + 1 = 2\alpha - n + 1. \quad (14)$$

Wide windows are therefore the signature of a hypergraph with large triangle-free sets — a sparse one.

For $u(N)$ we can write the critical sets down. Let

$$A_s = \{\text{all } p \text{ flavours of } E_{ij}, i < j\} \cup \{s_i \text{ flavours of } E_{ii}\}, \quad \max(0, p - q + 1) \leq s_i \leq \min(p, q - 1). \quad (15)$$

Each A_s is triangle-free: its strictly upper part admits no closed loop, by the index-loop argument of §2, and site i holds only $s_i \leq q - 1$ flavours, one short of the q that a self-loop needs. Its complement — the strictly lower part, acyclic for the same reason, together with $p - s_i \leq q - 1$ flavours per site — is triangle-free by the identical argument. Every A_s is thus critical, and as the s_i range over their allowed values $\sum_i s_i$ realises *every* integer in a range of length $N(2 \min(p, q - 1) - p)$. Hence

$$W \geq N(2 \min(p, q - 1) - p) + 1 \quad (16)$$

which at $p = 1$ reads $W \geq N + 1$, i.e. the window $[\frac{N(N-1)}{2}, \frac{N(N+1)}{2}]$ — **exactly Chen’s single-matrix result, recovered combinatorially with no Casimir argument and no appeal to solvability.**

The no-go follows immediately. For $p \leq q - 1$ we have $W \geq pN + 1$, so concentration fails for every $N > (q - 1)/p$; at Chen’s values $p = q = 3$ this reads $W \geq N + 1 > q$ for all $N \geq 3$. The failure is not marginal and it does not improve with N — it gets worse linearly.

4 Why gauge invariance is responsible

4.1 Positive roots carry no terms

The bound (14) is driven entirely by α , the largest triangle-free set, so the question becomes why gauge theories have such large ones. The answer is a single observation about weights. Gauge invariance means each term of Q is a singlet, so the weights of its q modes sum to zero. Inside $\mathfrak{n}^+ \otimes \mathbb{C}^p$ — all flavours of all positive-root vectors — every mode carries a positive root, and a non-empty sum of positive roots is a positive combination of simple roots, which is never zero. **No triangle fits inside the positive-root subspace**, for any reductive $\mathfrak{g}$ and any invariant coupling whatsoever.

Since $|\Delta^+| = \frac{1}{2}(\dim \mathfrak{g} - r)$, this alone gives

$$\alpha \geq p|\Delta^+| = \frac{1}{2}(n - pr), \quad (17)$$

so half of the modes of any gauge theory are automatically triangle-free. Writing α_{cart} for the largest triangle-free subset of the Cartan sector and feeding this into (14), the $|\Delta^+|$ cancels and

$$W \geq 2\alpha_{\text{cart}} - pr + 1, \quad (18)$$

so the window bound depends only on the rank and on what happens in the Cartan — the root sector contributes a fixed half and drops out. For $u(N)$ the Cartan piece is easy: a trace of q diagonal matrices localises, $\text{Tr}[H_1 \cdots H_q] = \sum_i \prod_t (H_t)_{ii}$, so a Cartan triangle needs q distinct flavours at a single site and $\alpha_{\text{cart}} = \min(p, q - 1)N$, which recovers (16). We have found (18) tight in all six measured cases.

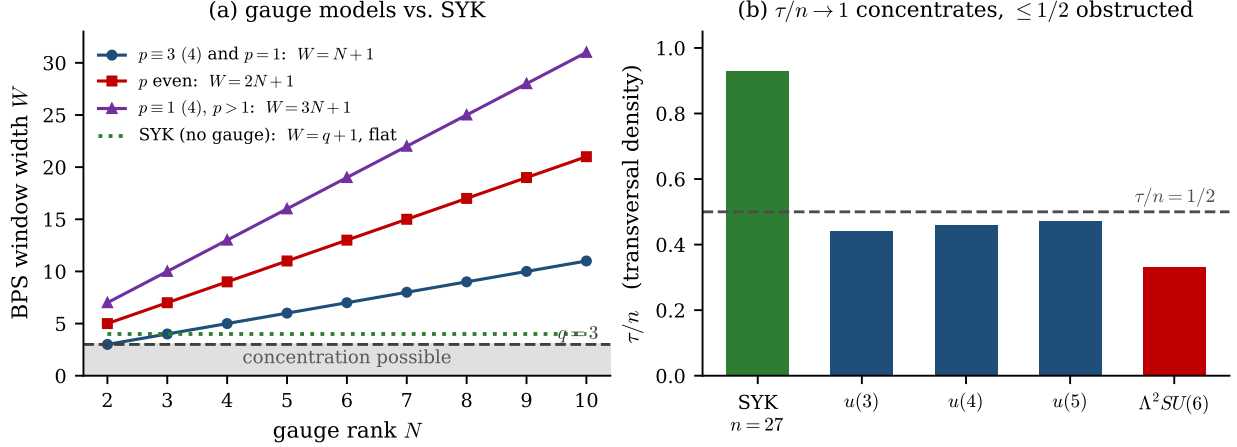


Figure 3: (a) Window width against gauge rank. The grading period $q = 3$ is fixed, so concentration is possible only inside the shaded band; every gauge model leaves it by $N = 3$, while SYK stays flat at $q + 1$. (b) The transversal density τ/n : SYK approaches 1 (no triangle-free set larger than $q - 1$), gauge models sit near $\frac{1}{2}$ (the Borel half), and Λ^2 matter is worse still.

4.2 SYK is not obstructed

The contrast with SYK is stark, and it is visible in the same quantity. With no gauge symmetry and generic q -local couplings, *every* q -subset of the n fermions is a triangle. A triangle-free set can then have at most $q - 1$ elements, so $\alpha_{\text{SYK}} = q - 1$ and the estimate (14) reads $W \geq 2(q - 1) - n + 1$, which is negative as soon as $n \geq 2q - 1$. The bound is simply **vacuous**: the argument of §3 imposes no obstruction at all on SYK, which is consistent with the result of Chang, Chen, Sia and Yang that $\mathcal{N} = 2$ SYK does concentrate, with a window of width $O(q)$ independent of n .

4.3 The pinpoint

Everything reduces to one comparison:

$$\alpha_{\text{SYK}} = q - 1 = O(1) \quad \text{versus} \quad \alpha_{\text{gauge}} \geq \frac{1}{2}(n - pr) = \Theta(n), \quad (19)$$

and the entire difference is the Borel subalgebra. The measured transversal densities in Fig. 3b make the contrast concrete: $\tau/n = 0.83, 0.89, 0.93$ for SYK at $n = 12, 18, 27$, against $0.44, 0.46, 0.47$ for $u(3), u(4), u(5)$ at $p = 3$. The fraction of triples that gauge invariance permits falls like $6/N^3$ for adjoint matter, while for SYK it is 1.

This excludes three candidate explanations one might otherwise reach for. It is not the flavour count, since (16) holds for every p and the parity pattern in w_s never brings W below $N + 1$. It is not randomness of the couplings: at $p = q = 3$ the coupling space is a single point, and in general generic couplings are the *best* case, because matrix ranks are lower semicontinuous in ω , so specialising can only enlarge the cohomology. And it is not the loss of Casimir solvability at $p = 3$, which is a symptom rather than a cause — indeed (16) reproduces the $p = 1$ solvable answer without ever using solvability. What remains is gauge invariance itself: it forces weight-zero terms, and half of a reductive Lie algebra admits none.

There is a dichotomy lurking here that is worth stating plainly, because it explains why no amount of tuning inside this class will help. Since $w_s \geq 2$ always, a window that stays bounded as $r \rightarrow \infty$ would require $w_s = 1$, which (12) forbids. One may fix the rank and grow p , in which case W does stay bounded — $u(2)$ with $p \equiv 3 \pmod{4}$ has $W = 3$ for every p , with extensive BPS entropy and a 1:2:1 profile — but such a family has no matrix large- N limit at all. Or one may grow the rank, obtaining the matrix limit but with $W \geq r + 1 \rightarrow \infty$. Both cannot hold at once. Concentration is a property of a *family* rather than of any single model, and for a matrix model the family parameter must be the rank.

Fortuity: embedding $\mathfrak{gl}(N) \hookrightarrow \mathfrak{gl}(N+1)$ leaves the new site empty

(a) rank 3 (BPS state)

s_i	p	p
0	s_i	p
0	0	s_i

(b) rank 4 embedded: LIFTED

s_i	p	p	0
0	s_i	p	0
0	0	s_i	0
0	0	0	0

new site: self-loop
entirely EMPTY
 $\Rightarrow Q$ creates it

Figure 4: Embedding $\mathfrak{gl}(N) \hookrightarrow \mathfrak{gl}(N+1)$ as the upper-left block leaves every mode carrying the new index unoccupied. The self-loop at the new site is then an entirely empty triangle, so Q_{N+1} creates it and the state is lifted.

5 Fortuity

5.1 Chen's argument, and why it does not suffice here

For the single-matrix model the BPS states fill the representation r_* , whose Young diagram is the staircase with $2N - 2i$ boxes in row i and exactly $N - 1$ rows. Going down in rank, that diagram has too many rows to exist as a $U(M)$ irrep; going up, it is no longer the maximal-Casimir representation. The states therefore exist at one rank only — they are *fortuitous* rather than monotone.

As noted in §2, the diagram of λ^* has rows $p(2N - 2i)$, i.e. p times Chen's staircase with the same $N - 1$ rows, so adding flavours stretches the rows without producing any new ones and his argument generalises verbatim to all p . The difficulty is elsewhere: unlike the single-matrix case, our BPS states are *not* confined to one irrep. At $N = 3$, $p = 3$ all thirteen irreps with $C_2 \geq 24$ carry BPS states, each with window width 4. A representation-theoretic argument about λ^* alone therefore cannot settle fortuity for this family, and we give a local argument instead — one that works state by state and does not care which irrep the state sits in.

5.2 The stabilisation map vanishes

Embed $\mathfrak{gl}(N) \hookrightarrow \mathfrak{gl}(N+1)$ as the upper-left block. Traces of products of block matrices restrict correctly, so this is a chain map and induces a stabilisation map $H_N \rightarrow H_{N+1}$ on cohomology. The claim is that this map is identically zero whenever $p \geq q$, so that no BPS state is monotone.

The reason is visible in Fig. 4. Under the embedding, every mode carrying the index $N + 1$ is unoccupied — there is nothing in the new row or column. Consider the self-loop at the new site, $T = \{\Psi_{N+1,N+1}^{a_1}, \dots, \Psi_{N+1,N+1}^{a_q}\}$ with q distinct flavours, which exists because $p \geq q$. It satisfies $T \cap S = \emptyset$ for any embedded state S , and a triangle disjoint from S is precisely one that Q can create: the Pauli blocking that made critical states closed in §3 is absent here. So $Q_{N+1}m_S \neq 0$, the state is not closed at rank $N + 1$, and it is lifted. For $p < q$ the same conclusion holds for the explicit sets (15), using the mixed loop $i \rightarrow N + 1 \rightarrow j \rightarrow i$ with $i < j$: the two edges through $N + 1$ are new and hence empty, and the return edge (j, i) is lower-triangular and hence empty as well. We have verified this at $p = q = 3$, where 9/9 BPS states of rank 2 and 27/27 of rank 3 are lifted, with the disjoint triangle exhibited explicitly in each case.

It is striking that this argument and the no-go of §3 rest on the same object — an empty triangle — yet pull in opposite directions. Enlarging the gauge group manufactures a fresh site whose self-loop is necessarily empty, and that *produces* fortuity; the very same enlargement leaves the Borel half of the algebra free of triangles, and that *destroys* concentration. The two properties one wants a fortuitous matrix model to have respond to the single available structural knob with opposite signs, which may be why the combination has proved hard to realise.

6 Status, and what is open

Established. The reduction (4) of the whole problem to multiplication by a form; the index-loop restriction and the resulting factorisation (10) with its width formula (11); the bound $w_s \geq 2$ and hence $W \geq r + 1$ at the maximal weight, (12); the closed and non-exact properties of critical states and the range (14); the explicit family (15) and the resulting (16) for $u(N)$, together with the no-go for $N > (q - 1)/p$; the positive-root observation and (18) for any reductive $\mathfrak{g}$; and the vanishing of the stabilisation map.

Verified numerically. All cohomology is computed by exact integer linear algebra — Gaussian elimination over $\mathbb{F}_p$, with every rank confirmed against a second prime — using the $U(N)$ weight decomposition to replace 2^{pN^2} by weight spaces of dimension at most 3×10^4 , and Kostant inversion to obtain per-irrep counts. Every number is cross-checked against the refined Witten index, against exact diagonalisation at $N = 2$, and against Chen’s closed form at $p = 1$.

Open.

- **$\mathfrak{su}(N)$.** The set (15) is *not* triangle-free there: the Cartan is $(N - 1)$ -dimensional and spread across sites, so Cartan triangles need not localise and the counting that gave α_{cart} for $u(N)$ fails. The true transversal numbers are $\tau = 12, 21, 36$ for $N = 3, 4, 5$ at $p = 3$, giving bounds 1, 4, 1 against measured widths 3, 4, 5 — valid, tight only at $N = 4$, and vacuous at odd N , where $\alpha = n/2$ exactly. The factorisation (10) also fails, the measured profiles 13, 81, 81, 13 and 15, 258, 486, 258, 15 being manifestly non-binomial. A sharp $\mathfrak{su}(N)$ statement needs α_{cart} in closed form, or classes that are not single monomials.
- **An upper bound on W .** Everything above bounds the window from below. The complementary statement — that the cohomology *vanishes* outside a band, equivalently that $\omega^\wedge$ is injective in low degrees — is verified in six models, three of them over complete spectra, but has not been established. It is not implied by the critical-set argument, which constructs classes rather than excluding them. It is the more valuable of the two open items: with it, the closed-form W at the maximal weight would fix the window for every irrep and the no-go would become spectrum-wide.
- **Non-adjoint matter.** Fundamental matter forces $q = N$ (the baryon) but has $\tau = p$ independent of N , hence $W \geq p(N - 2) + 1$: worse, not better. Two-index antisymmetric matter at $SU(6)$ gives $\tau/n = 0.33$, also worse. A general statement across representations is open.
- **Chaos.** Nothing here bears on whether these models are chaotic. The three-flavour H is not a Casimir function, so a super-Schwarzian near-BPS spectrum remains possible without concentration.

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